

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

Reinforcement Learning

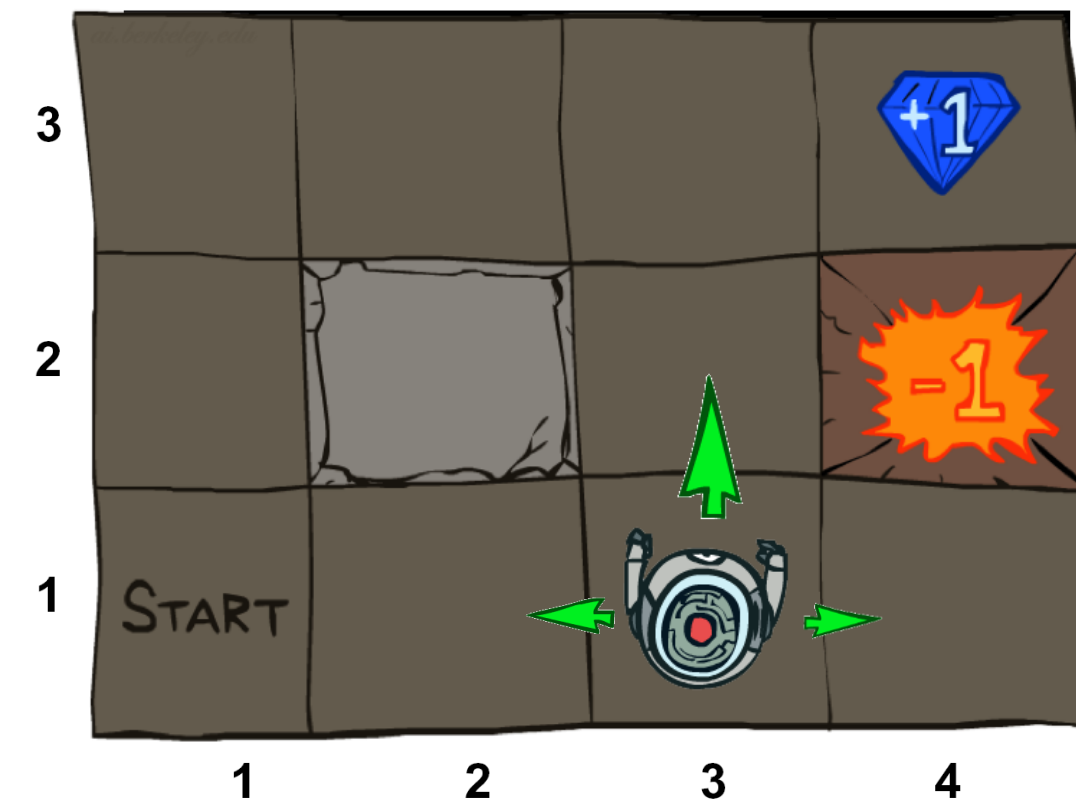
Texas A&M University

Topics

- What is reinforcement learning?
- An example application and Markov Decision Process
- Value Iteration
- Q Learning (and Policy Iteration)

Example: Grid World

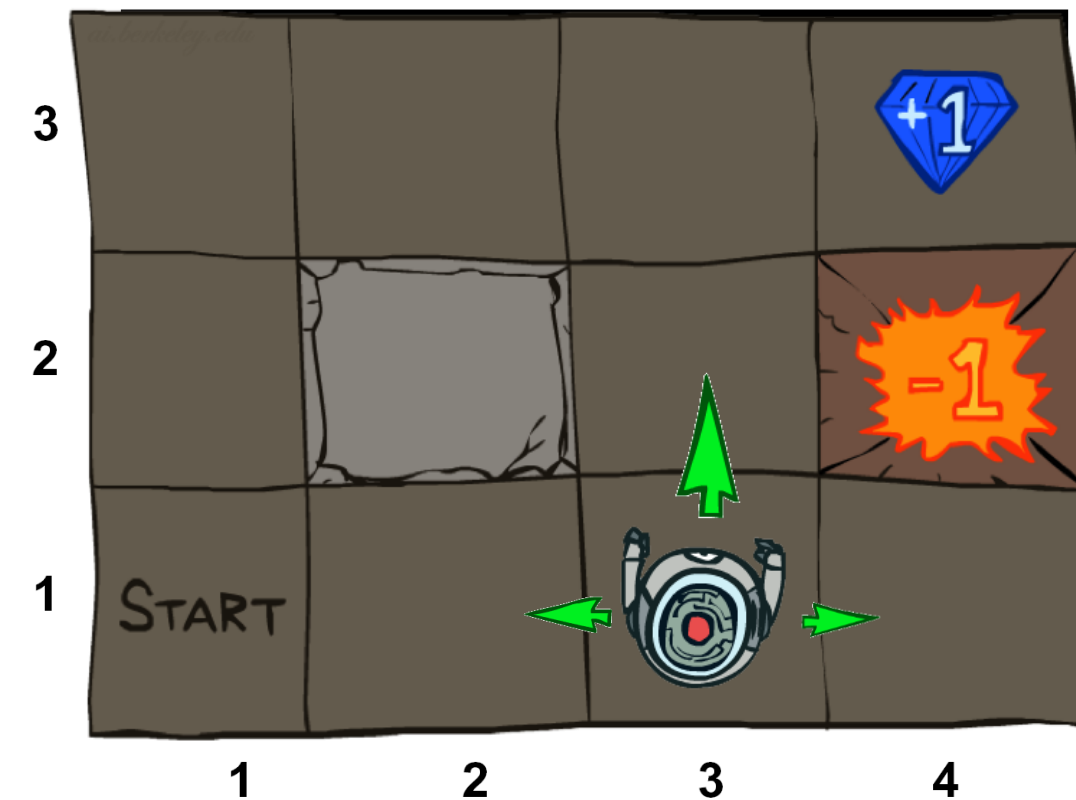
- A maze-like problem
 - The agent lives in a grid
 - Walls block the agent's path
- Noisy movement: actions do not always go as planned
 - 80% of the time, the action North takes the agent North (if there is no wall there)
 - 10% of the time, North takes the agent West; 10% East
 - If there is a wall in the direction the agent would have been taken, the agent stays put
- The agent receives rewards each time step
 - -0.1 per step (fuel loss)
 - +1 if arriving at (4,3) and -1 for arriving at (4,2)
- Goal: maximize accumulated rewards



Courtesy: Dr. Sharon

Markov Decision Processes

- An MDP is defined by:
 - A set of states $\mathcal{S}$
 - A set of actions $\mathcal{A}$
 - State-transition probabilities $P(s'|s, a)$
 - Probability of arriving to s' after performing a at s
 - Also called the model dynamics
 - A reward function $R(s, a, s')$
 - The utility gained from arriving to s' after performing a at s
 - Sometimes just $R(s, a)$ or even $R(s)$
 - A **start state**
 - Maybe a **terminal state**



Courtesy: Dr. Sharon

MDP formalization - Autonomous driving [“Learning to

Drive in a Day”, Kendall et. al., 2018]

- State:
 - image from front camera
- Actions:
 - Steering from $[-1,1]$, acceleration from $[0,1]$, brake from $[0,1]$
- Reward:
 - Drive off road = -10
 - Hit a person = -1000
 - Drive at speed limit = +2
 - Align with lane = +4
 - Run a stop sign = -5
 - Erratic driving = -3
 - ...
- State-transition probabilities:
 - defined by stochasticity in action outcomes and other actuators

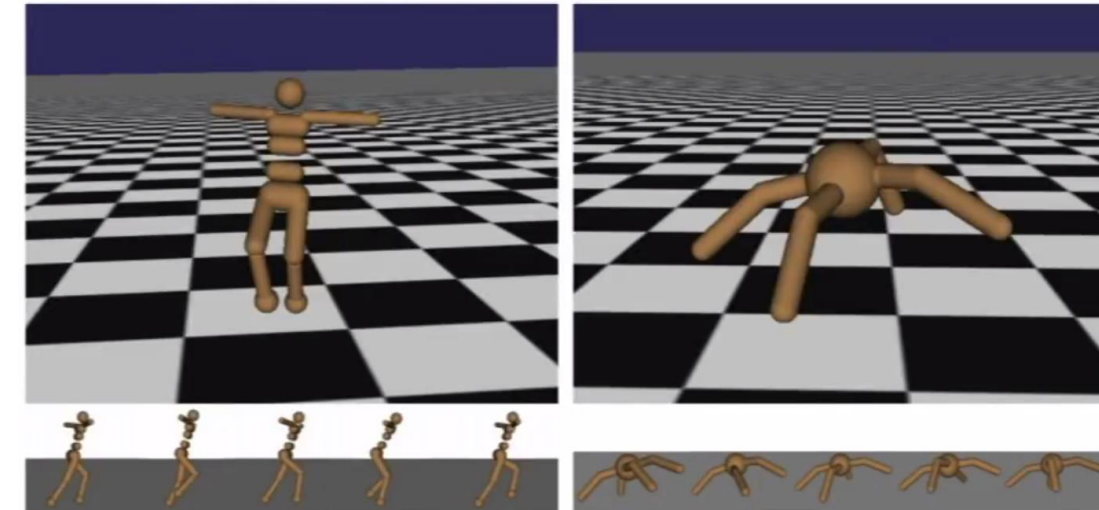


Courtesy: Dr. Sharon

MDP formalization - Robot locomotion

[“Policy gradient reinforcement learning for fast quadrupedal locomotion”, Kohl & Stone, 2004]

- State:
 - angle and position of joints, obstacles/pits
- Actions:
 - torques applied to joints
- Reward:
 - forward speed
- State-transition probabilities:
 - defined by stochasticity in action outcomes and obstacles

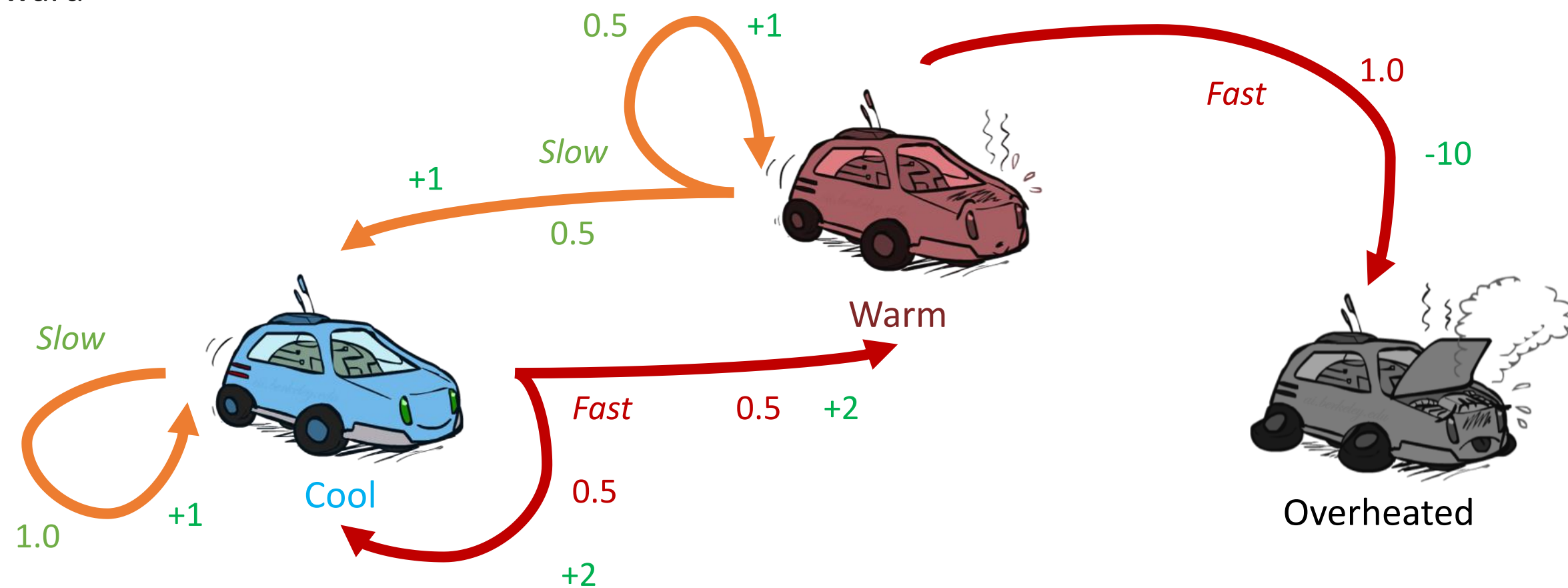


Courtesy: Dr. Sharon

Race car example

- A robot car wants to travel far, quickly
- Three states: **Cool**, **Warm**, Overheated
- Two actions: *Slow*, *Fast*
- Going faster gets double reward
- What is V^* ?
- What is π^* ?
- What's the problem?

$$\max_a \left[\sum_{s'} P(s'|s, a) (R(s, a, s') + v^*(s')) \right]$$



Courtesy: Dr. Sharon

Planning Algorithms

- Recall for planning algorithms - the environment model is known and all we want to do is find the optimal policy.
- This means that $\mathbb{E}[r(s, a)]$ and $\mathbb{P}[s'|s, a]$ are known.
- Do not need to learn environment, just find the best course of action.
- Three algorithms will be covered - Value Iteration, Policy Iteration, and Linear Programming